FAIRCHILD SEMICONDUCTOR™

February 1998

74F240 • 74F241 • 74F244
Octal Buffers/Line Drivers with 3-STATE Outputs

General Description

The 'F240, 'F241 and 'F244 are octal buffers and line drivers designed to be employed as memory and address drivers, clock drivers and bus-oriented transmitters/receivers which provide improved PC and board density.

Features

- 3-STATE outputs drive bus lines or buffer memory address registers
- Outputs sink 64 mA (48 mA mil)
- 12 mA source current
- Input clamp diodes limit high-speed termination effects
- Guaranteed 4000V minimum ESD protection

Ordering Code:

Commercial	Military	Package Number	Package Description
74F240PC		N20A	20-Lead (0.300" Wide) Molded Dual-In-Line
	54F240DM (Note 2)	J20A	20-Lead Ceramic Dual-In-Line
74F240SC (Note 1)		M20B	20-Lead (0.300" Wide) Molded Small Outline, JEDEC
74F240SJ (Note 1)		M20D	20-Lead (0.300" Wide) Molded Small Outline, EIAJ
	54F240FM (Note 2)	W20A	20-Lead Cerpack
	54F240LM (Note 2)	E20A	20-Lead Ceramic Leadless Chip Carrier, Type C
74F241PC		N20A	20-Lead (0.300" Wide) Molded Dual-In-Line
	54F241DM (Note 2)	J20A	20-Lead Ceramic Dual-In-Line
74F241SC (Note 1)		M20B	20-Lead (0.300" Wide) Molded Small Outline, JEDEC
74F241SJ (Note 1)		M20D	20-Lead (0.300" Wide) Molded Small Outline, EIAJ
	54F241FM (Note 2)	W20A	20-Lead Cerpack
	54F241LM (Note 2)	E20A	20-Lead Ceramic Leadless Chip Carrier, Type C
74F244PC		N20A	20-Lead (0.300" Wide) Molded Dual-In-Line
	54F244DM (Note 2)	J20A	20-Lead Ceramic Dual-In-Line
74F244SC (Note 1)		M20B	20-Lead (0.300" Wide) Molded Small Outline, JEDEC
74F244SJ (Note 1)		M20D	20-Lead (0.300" Wide) Molded Small Outline, EIAJ
74F244MSA (Note 1)		MSA20	20-Lead Molded Shrink Small Outline, EIAJ Type II
	54F244FM (Note 2)	W20A	20-Lead Cerpack
	54F244LM (Note 2)	E20A	20-Lead Ceramic Leadless Chip Carrier, Type C

Note 1: Devices also available in 13" reel. Use Suffix = SCX, SJX and MSAX.

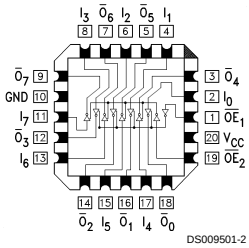
Note 2: Military grade device with environmental and burn-in processing. Use suffix = DMQB, FMQB and LMQB.

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Connection Diagrams

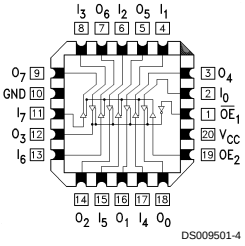
Pin Assignment for LCC

'F240



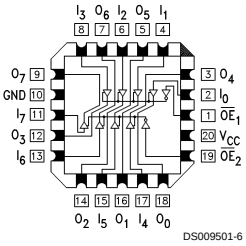
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'F241



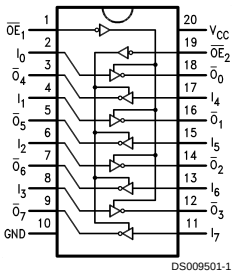
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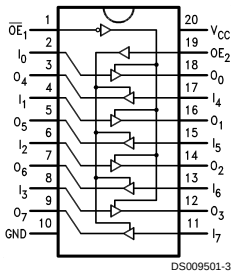


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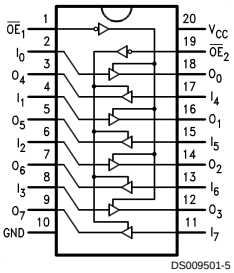
Pin Assignment for DIP, SOIC, SSOP and Flatpak



DS009501-1



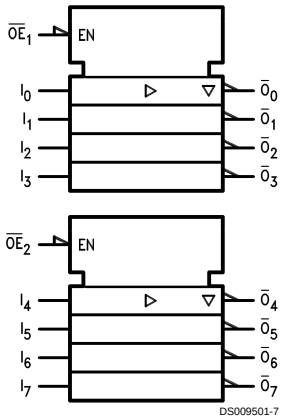
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DS009501-5

Logic Symbols

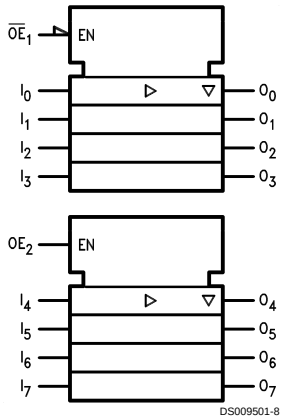
IEEE/IEC
'F240



DS009501-7

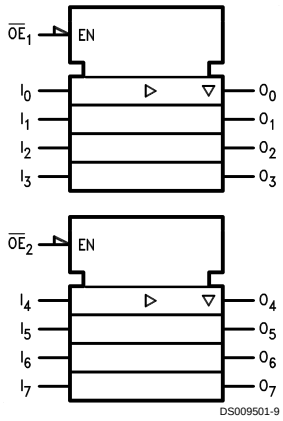
Logic Symbols (Continued)

IEEE/IEC
'F241



DS009501-8

IEEE/IEC
'F244



DS009501-9

Unit Loading/Fan Out

Pin Names	Description	54F/74F	
		U.L. HIGH/LOW	Input I_{IH}/I_{IL} Output I_{OH}/I_{OL}
$\overline{OE}_1$, $\overline{OE}_2$	3-STATE Output Enable Input (Active LOW)	1.0/1.667	20 µA/−1 mA
OE_2	3-STATE Output Enable Input (Active HIGH)	1.0/1.667	20 µA/−1 mA
I_0–I_7	Inputs ('F240)	1.0/1.667 (Note 3)	20 µA/−1 mA
I_0–I_7	Inputs ('F241, 'F244)	1.0/2.667 (Note 3)	20 µA/−1.6 mA
$\overline{O}_0$–$\overline{O}_7$, O_0–O_7	Outputs	600/106.6 (80)	−12 mA/64 mA (48 mA)

Note 3: Worst-case 'F240 enabled; 'F241, 'F244 disabled

Truth Tables

'F240

$\overline{OE}_1$	D_{1n}	O_{1n}	$\overline{OE}_2$	D_{2n}	O_{2n}
H	X	Z	H	X	Z
L	H	L	L	H	L
L	L	H	L	L	H

'F241

$\overline{OE}_1$	D_{1n}	O_{1n}	OE_2	D_{2n}	O_{2n}
H	X	Z	L	X	Z
L	H	H	H	H	H
L	L	L	H	L	L

'F244

$\overline{OE}_1$	D_{1n}	O_{1n}	$\overline{OE}_2$	D_{2n}	O_{2n}
H	X	Z	H	X	Z
L	H	H	L	H	H
L	L	L	L	L	L

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial
Z = High Impedance

Absolute Maximum Ratings (Note 4)

Storage Temperature	−65˚C to +150˚C
Ambient Temperature under Bias	−55˚C to +125˚C
Junction Temperature under Bias	−55˚C to +175˚C
Plastic	−55˚C to +150˚C
V_{CC} Pin Potential to Ground Pin	−0.5V to +7.0V
Input Voltage (Note 5)	−0.5V to +7.0V
Input Current (Note 5)	−30 mA to +5.0 mA
Voltage Applied to Output in HIGH State (with V_{CC} = 0V)	
Standard Output	−0.5V to V_{CC}
3-STATE Output	−0.5V to +5.5V
Current Applied to Output in LOW State (Max)	twice the rated I_{OL} (mA)
ESD Last Passing Voltage (Min)	4000V

Recommended Operating Conditions

Free Air Ambient Temperature	
Military	−55˚C to +125˚C
Commercial	0˚C to +70˚C
Supply Voltage	
Military	+4.5V to +5.5V
Commercial	+4.5V to +5.5V

Note 4: Absolute maximum ratings are values beyond which the device may be damaged or have its useful life impaired. Functional operation under these conditions is not implied.

Note 5: Either voltage limit or current limit is sufficient to protect inputs.

DC Electrical Characteristics

Symbol	Parameter		54F/74F			Units	V_{CC}	Conditions
			Min	Typ	Max			
V_{IH}	Input HIGH Voltage		2.0			V		Recognized as a HIGH Signal
V_{IL}	Input LOW Voltage				0.8	V		Recognized as a LOW Signal
V_{CD}	Input Clamp Diode Voltage				−1.2	V	Min	I_{IN} = −18 mA
V_{OH}	Output HIGH Voltage	54F 10% V_{CC}	2.4					I_{OH} = −3 mA
		54F 10% V_{CC}	2.0					I_{OH} = −12 mA
		74F 10% V_{CC}	2.4			V	Min	I_{OH} = −3 mA
		74F 10% V_{CC}	2.0					I_{OH} = −15 mA
		74F 5% V_{CC}	2.7					I_{OH} = −3 mA
V_{OL}	Output LOW Voltage	54F 10% V_{CC}			0.55	V	Min	I_{OL} = 48 mA
		74F 10% V_{CC}			0.55			I_{OL} = 64 mA
I_{IH}	Input HIGH Current	54F			20.0	µA	Max	V_{IN} = 2.7V
		74F			5.0			
I_{BVI}	Input HIGH Current Breakdown Test	54F			100	µA	Max	V_{IN} = 7.0V
		74F			7.0			
I_{CEX}	Output HIGH Leakage Current	54F			250	µA	Max	V_{OUT} = V_{CC}
		74F			50			
V_{ID}	Input Leakage Test	74F	4.75			V	0.0	I_{ID} = 1.9 µA All Other Pins Grounded
I_{OD}	Output Leakage Circuit Current	74F			3.75	µA	0.0	V_{IOD} = 150 mV All Other Pins Grounded
I_{IL}	Input LOW Current				−1.0	mA	Max	V_{IN} = 0.5V ($\overline{OE}_1$, $\overline{OE}_2$, OE_2, D_n ('F240))
					−1.6			V_{IN} = 0.5V (D_n ('F241, 'F244))
I_{OZH}	Output Leakage Current				50	µA	Max	V_{OUT} = 2.7V
I_{OZL}	Output Leakage Current				−50	µA	Max	V_{OUT} = 0.5V
I_{OS}	Output Short-Circuit Current		−100		−225	mA	Max	V_{OUT} = 0V
I_{ZZ}	Bus Drainage Test				500	µA	0.0V	V_{OUT} = 5.25V
I_{CCH}	Power Supply Current ('F240)			19	29	mA	Max	V_O = HIGH
I_{CCL}	Power Supply Current ('F240)			50	75	mA	Max	V_O = LOW
I_{CCZ}	Power Supply Current ('F240)			42	63	mA	Max	V_O = HIGH Z
I_{CCH}	Power Supply Current ('F241, 'F244)			40	60	mA	Max	V_O = HIGH
I_{CCL}	Power Supply Current ('F241, 'F244)			60	90	mA	Max	V_O = LOW

DC Electrical Characteristics (Continued)

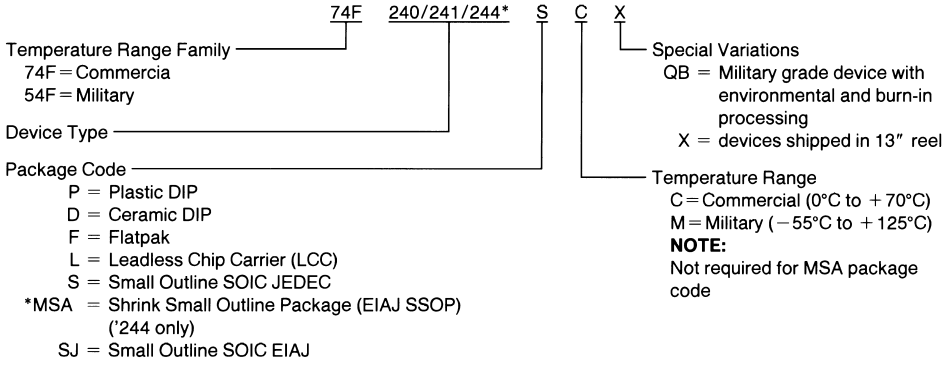
Symbol	Parameter	54F/74F			Units	V_{CC}	Conditions
		Min	Typ	Max			
I_{CCZ}	Power Supply Current ('F241, 'F244)		60	90	mA	Max	V_O = HIGH Z

AC Electrical Characteristics

Symbol	Parameter	74F T_A = +25°C, V_{CC} = +5.0V, C_L = 50 pF			54F T_A, V_{CC} = Mil, C_L = 50 pF		74F T_A, V_{CC} = Com, C_L = 50 pF		Units
		Min	Typ	Max	Min	Max	Min	Max	
t_{PLH}	Propagation Delay	3.0	5.1	7.0	3.0	9.0	3.0	8.0	ns
t_{PHL}	Data to Output ('F240)	2.0	3.5	4.7	2.0	6.0	2.0	5.7	
t_{PZH}	Output Enable Time ('F240)	2.0	3.5	4.7	2.0	6.5	2.0	5.7	
t_{PZL}		4.0	6.9	9.0	4.0	10.5	4.0	10.0	ns
t_{PHZ}	Output Disable Time ('F240)	2.0	4.0	5.3	2.0	6.5	2.0	6.3	
t_{PLZ}		2.0	6.0	8.0	2.0	12.5	2.0	9.5	
t_{PLH}	Propagation Delay	2.5	4.0	5.2	2.0	6.5	2.5	6.2	ns
t_{PHL}	Data to Output ('F241, 'F244)	2.5	4.0	5.2	2.0	7.0	2.5	6.5	
t_{PZH}	Output Enable Time	2.0	4.3	5.7	2.0	7.0	2.0	6.7	
t_{PZL}	('F241, 'F244)	2.0	5.4	7.0	2.0	8.5	2.0	8.0	ns
t_{PHZ}	Output Disable Time	2.0	4.5	6.0	2.0	7.0	2.0	7.0	
t_{PLZ}	('F241, 'F244)	2.0	4.5	6.0	2.0	7.5	2.0	7.0	

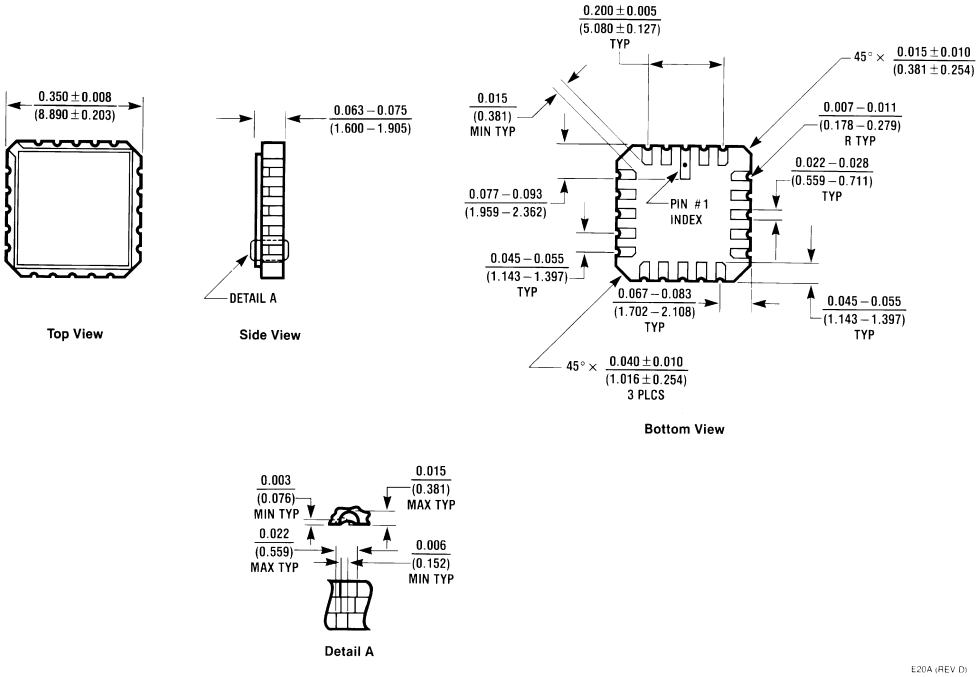
Ordering Information

The device number is used to form part of a simplified purchasing code where the package type and temperature range are defined as follows:

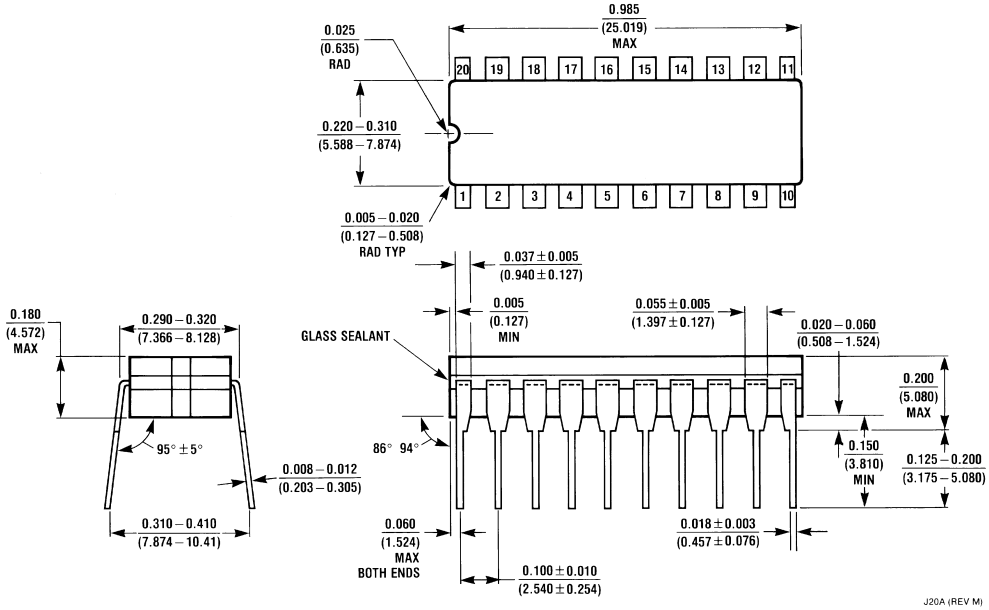


DS009501-10

Physical Dimensions inches (millimeters) unless otherwise noted

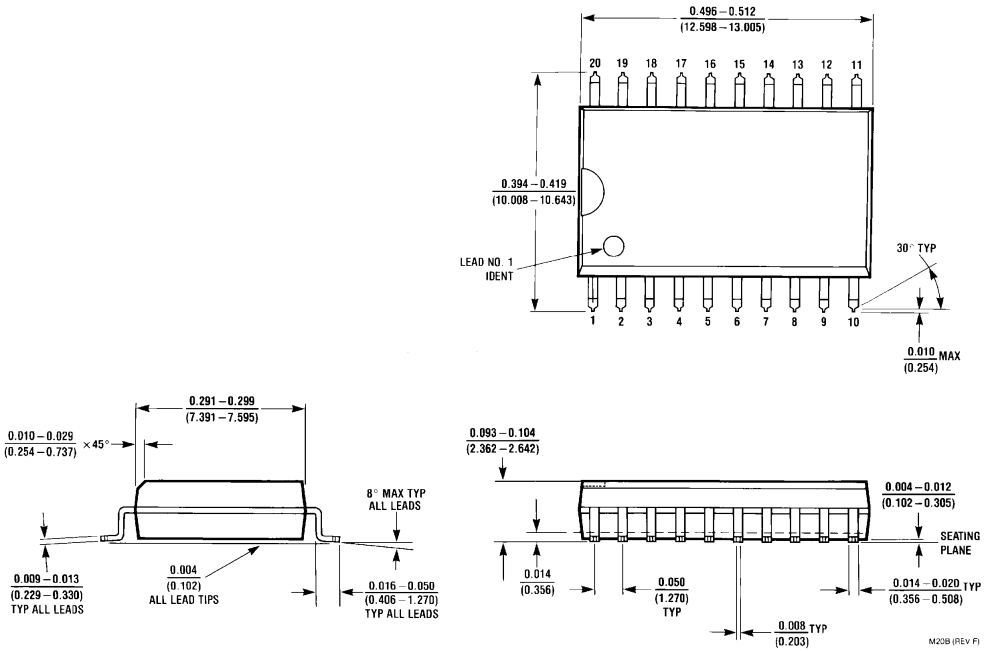


20-Lead Ceramic Leadless Chip Carrier (L)
Package Number E20A

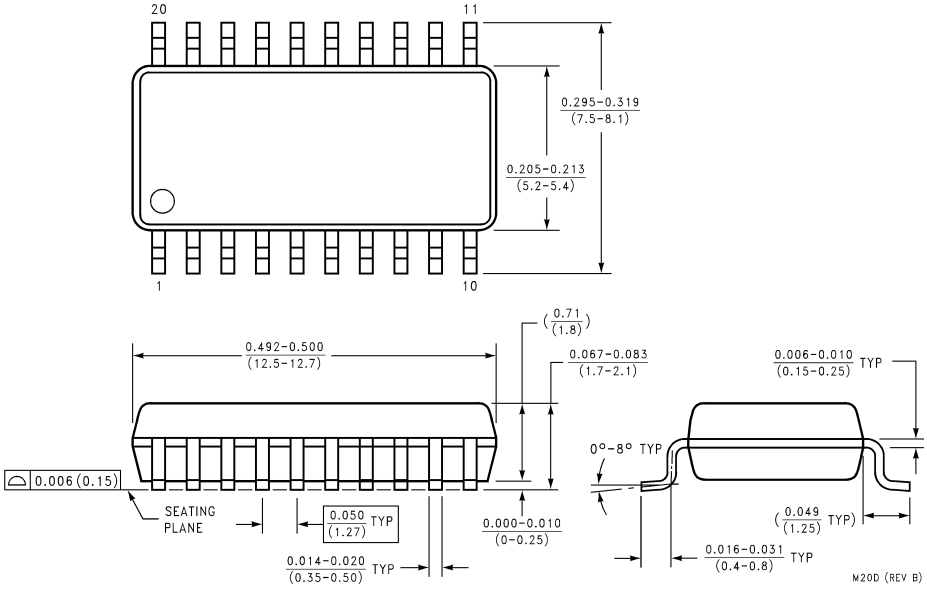


20-Lead Ceramic Dual-In-Line Package (D)
Package Number J20A

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)

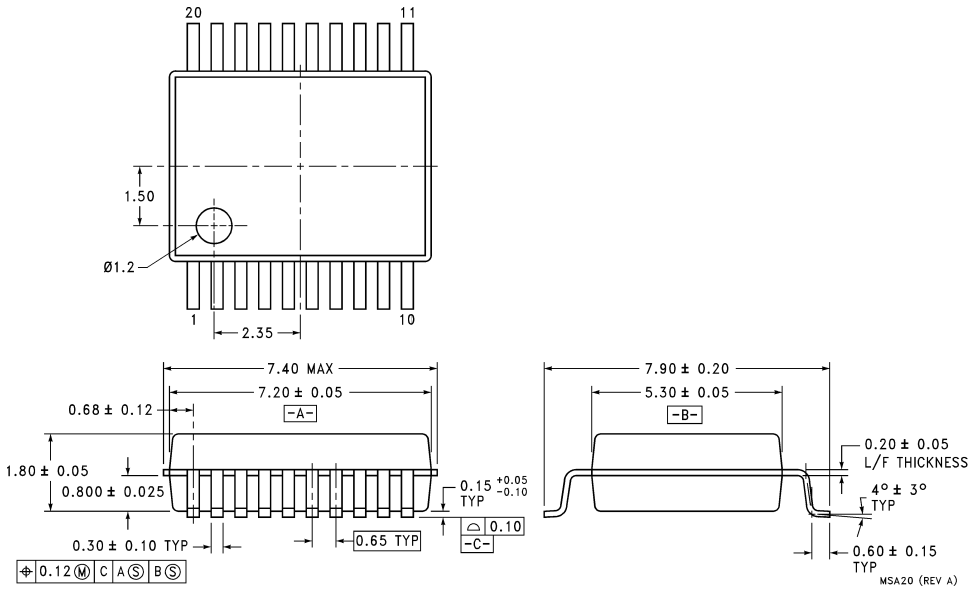


20-Lead (0.300" Wide) Molded Small Outline Package, JEDEC (S)
Package Number M20B

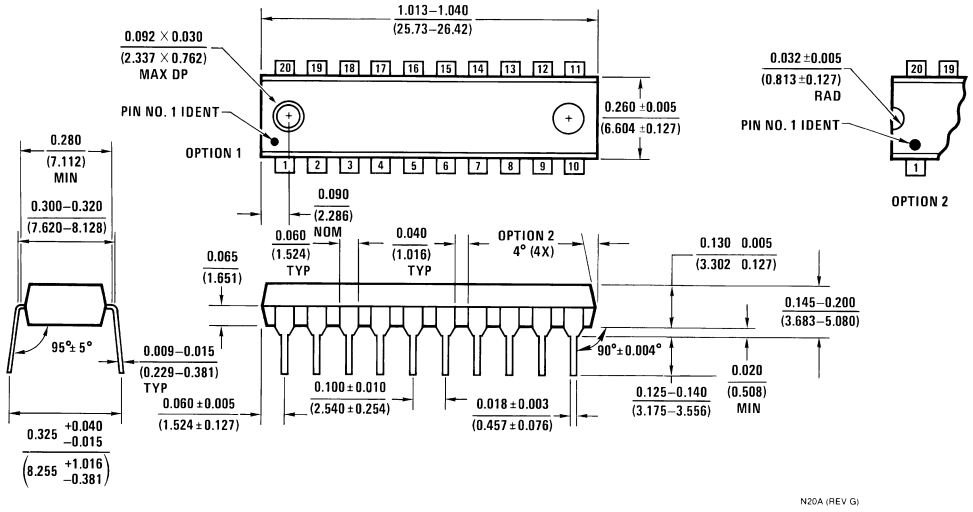


20-Lead (0.300" Wide) Molded Small Outline Package, EIAJ (SJ)
Package Number M20D

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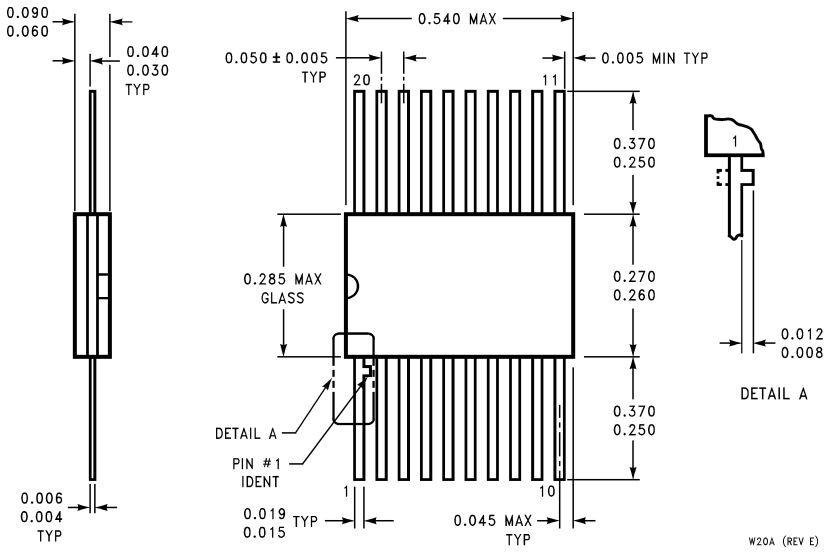


20-Lead Molded Shrink Small Outline, EIAJ Type II (MSA)
Package Number MSA20



20-Lead (0.300" Wide) Molded Dual-In-Line Package (P)
Package Number N20A

Physical Dimensions inches (millimeters) unless otherwise noted (Continued)



20-Lead Ceramic Flatpak (F)
Package Number W20A

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